

Distality in ordered abelian groups

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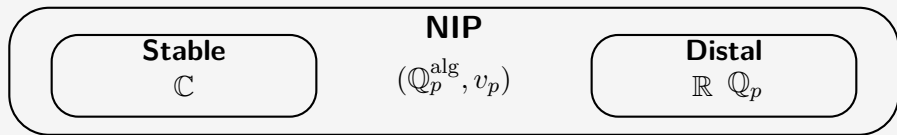
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Distality is the opposite of stability (among NIP theories).



Definition ([Sim13])

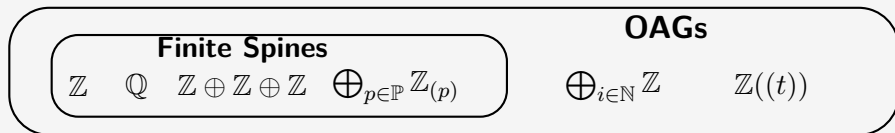
A theory is *distal* if for any sequence $(\bar{a}_i)_{i \in \mathbb{Q}}$ and any tuple $\bar{b}$, if

- $(\bar{a}_i)_{i \in \mathbb{Q}}$ is indiscernible and
- $(\bar{a}_i)_{i \in \mathbb{Q} \setminus \{0\}}$ is indiscernible over $\bar{b}$,

then $(\bar{a}_i)_{i \in \mathbb{Q}}$ is indiscernible over $\bar{b}$.

Roughly speaking, distal theories are those NIP theories which do not have a stable part.

Though all ordered abelian groups (OAGs) have NIP [GS84], only a distality criterion for OAGs with finite spines was known [Asc+22].



We have

- given a distality criterion for *all* OAGs.
- shown all OAGs have distal expansions.

Fix an OAG G . For convex subgroups H_1, H_2 , we write $H_1 \leq H_2 \iff H_1 \subseteq H_2$.

Definition

For $n \geq 2$ and $a \in G \setminus nG$, let $\mathfrak{s}_n(a)$ be the largest convex subgroup H such that $a \notin H + nG$. If $a \in nG$, we set $\mathfrak{s}_n(a) = \emptyset$.

An important fact is that we may regard $\mathfrak{s}_n$ as a valuation on G .

Lemma

- 1 $\mathfrak{s}_n(-a) = \mathfrak{s}_n(a)$.
- 2 $\mathfrak{s}_n(a + b) \leq \max \{ \mathfrak{s}_n(a), \mathfrak{s}_n(b) \}$ (*ultrametric inequality*).

We view $(G, \mathfrak{s}_n)$ in an analogue to valued fields.

Valued field	OAG
v	$(\mathfrak{s}_n)_{n \geq 2}$
Value group	spines
Residue field	ribs

Definition

We define the *spine* $\mathcal{S}_n = \text{Im}(\mathfrak{s}_n)$.

Theorem (Aschenbrenner, Chernikov, Gehret, and Ziegler [Asc+22])

Suppose that $|\mathcal{S}_p| < \infty$ for every prime p . Then G is distal iff $|G/pG| < \infty$ for all primes p .

Definition

For $H \in \mathcal{S}_n \setminus \{\emptyset\}$, define $G_{\leq H}^{\mathfrak{s}_n} = \{x \in G \mid \mathfrak{s}_n(x) \leq H\}$ and $G_{< H}^{\mathfrak{s}_n}$. We call $R_H^{\mathfrak{s}_n} = G_{\leq H}^{\mathfrak{s}_n} / G_{< H}^{\mathfrak{s}_n}$ the *rib* at H .

Finite Spines

$$\mathbb{Z} \quad \mathbb{Q} \quad \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z} \quad \bigoplus_{p \in \mathbb{P}} \mathbb{Z}_{(p)}$$

OAGs

$$\bigoplus_{i \in \mathbb{N}} \mathbb{Z} \quad \mathbb{Z}((t))$$

Theorem (O. [Oku26])

G is distal iff for every prime p , there is $N_p \in \mathbb{N}$ such that $|R_H^{s_p}| < N_p$ for every $H \in \mathcal{S}_p \setminus \{\emptyset\}$.

While some OAGs are not distal, we have also shown:

Theorem (O. [Oku26])

Every OAG has a distal expansion.

- [Asc+22] Matthias Aschenbrenner, Artem Chernikov, Allen Gehret, and Martin Ziegler. “Distality in valued fields and related structures”. *Transactions of the American Mathematical Society* 375.7 (2022), pp. 4641–4710 (cit. on pp. 3, 5).
- [GS84] Yuri Gurevich and Peter H. Schmitt. “The theory of ordered abelian groups does not have the independence property”. *Transactions of the American Mathematical Society* 284.1 (1984), pp. 171–182 (cit. on p. 3).
- [Oku26] Koki Okura. *Distality in ordered abelian groups*. 2026. arXiv: 2602.23256 [math.LO] (cit. on p. 6).
- [Sim13] Pierre Simon. “Distal and non-distal NIP theories”. *Annals of Pure and Applied Logic* 164.3 (2013), pp. 294–318 (cit. on p. 2).